Homework 38: Constrained Optimization in a Nutshell

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BCS A, E, F, G

Problem 1. ***Positive Definite Matrices and Quadratic Forms.*** Decide for or against the positive definiteness of these matrices, and write out the corresponding $f = \boldsymbol{x}^T A \boldsymbol{x}$:

a. $\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$. **b.** $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. **c.** $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$. **d.** $\begin{bmatrix} -1 & 2 \\ 2 & 8 \end{bmatrix}$.

Problem 2. ***Checking Inverses as Positive Definite Matrices.*** If $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is positive definite, test $A^{-1} = \begin{bmatrix} p & q \\ q & r \end{bmatrix}$ for positive definiteness.

Problem 3. ***Using a Change of Variables.*** Find the maximum and minimum values of $z = 4xy$ subject to the constraint $9x^2 + 4y^2 = 36$. Note that, you can't use the Constrained Optimization Theorem directly because the constraint is not $\|\boldsymbol{x}\|^2 = 1$. You have to make a change of variables first in order to transform the constraint to $\|\boldsymbol{x}\|^2 = 1$, and then use the method discussed within lectures.

Theorem 1. Let A be a symmetric matrix and define m and M as

$$m = \min\{\boldsymbol{x}^T A \boldsymbol{x} : \|\boldsymbol{x}\| = 1\}, \qquad M = \max\{\boldsymbol{x}^T A \boldsymbol{x} : \|\boldsymbol{x}\| = 1\}.$$

Then M is the greatest eigenvalue λ_1 of A and m is the least eigenvalue of A. The value of $\boldsymbol{x}^T A \boldsymbol{x}$ is M when $\boldsymbol{x}$ is a unit eigenvector $\boldsymbol{u}_1$ corresponding to M. The value of $\boldsymbol{x}^T A \boldsymbol{x}$ is m when $\boldsymbol{x}$ is a unit eigenvector corresponding to m.

Theorem 2. Let A, λ_1, and $\boldsymbol{u}_1$ be as in Theorem 1. Then the maximum value of $\boldsymbol{x}^T A \boldsymbol{x}$ subject to the constraints

$$\boldsymbol{x}^T \boldsymbol{x} = 1, \qquad \boldsymbol{x}^T \boldsymbol{u}_1 = 0,$$

is the second greatest eigenvalue, λ_2, and this maximum is attained when $\boldsymbol{x}$ is an eigenvector $\boldsymbol{u}_2$ corresponding to λ_2.

Problem 4. ***Computations With Additional Constraints.*** Let

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix},$$

and let $\boldsymbol{u}_1$ be a unit eigenvector corresponding to the greatest eigenvalue of A. Find the maximum value of $\boldsymbol{x}^T A \boldsymbol{x}$ subject to the conditions $\boldsymbol{x}^T \boldsymbol{x} = 1,\ \boldsymbol{x}^T \boldsymbol{u}_1 = 0$.

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Good Luck